

A Typed-Decision Model Matches 7B Language Models for Speech-Neuroprosthesis Rescoring

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Abstract

A speech neuroprosthesis decodes attempted speech from brain activity and ends by rescoring the decoder’s candidate sentences with a language model of several billion parameters, the only component that needs a GPU. Replacing that model with a cheaper one is hard: general language models asked to pick one sentence from a list answer from where a label sits in the list rather than from the sentence itself. We pose rescoring as a single typed decision, one call that returns a probability for every candidate, served by Jev, a hosted model trained for calibrated decisions, and combine it with the decoder’s own score. On 978 held-out sentences from a participant with ALS, where the published decoder alone reaches 8.1% word error, Jev reaches 7.5% against 7.8% for both OPT-6.7b and Qwen2.5-7B; with the decoder’s weight re-tuned, 6.9% against 7.2% and 7.4%. Jev is ahead in all four comparisons and at most 0.2 points behind at the 95% bound. It costs \$0.07 per thousand sentences and needs no GPU; a dedicated GPU running a 7B model is cheaper per sentence only above 43% utilisation, far beyond what one user generates. End-to-end latency over the internet is 262 ms, of which 62 ms is spent at the provider, the same order as a 7B model on a local GPU (27 ms) but not faster.

1 Introduction

A speech neuroprosthesis turns intracortical activity into text in three stages. A recurrent network trained with connectionist temporal classification [5] emits phoneme probabilities, a lexicon- and n -gram-constrained decoder turns them into a ranked list of candidate sentences, and a language model rescores that list before the winner is spoken [1, 2]. The Brain-to-Text benchmark’s reference implementation rescores with OPT-6.7b [4, 3]. This last stage is the only one that needs a multi-billion-parameter model, so it sets the accelerator, the power budget and the cost of the system.

The rescorer’s job is selection: the answer is one of N sentences already written down. A model that only selects should suffice, and one call over the whole list is the natural shape for it. General language models handle that shape badly. Asked to choose from a list, they favour positions over contents [10, 11], and in this task open models posed this way place most of their answers on the decoder’s own top candidate and almost never pick the truth when they deviate from it (Section 5.2).

A typed-decision model is trained for this output type. Given a state and a question with a fixed set of answers, it returns a probability for each answer in one pass and cannot produce anything outside the set. Jev [13] is such a model, trained with reinforcement learning for calibrated decisions and served through an API. Early third-party evaluations report lower latency

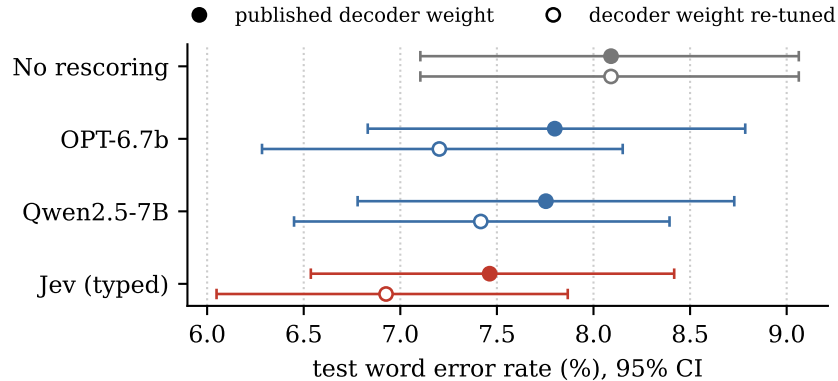


Figure 1: Test word error with 95% bootstrap intervals, 978 sentences, identical candidate lists. Marginal intervals overlap because they share sentence-level noise; the paired tests in Tables 1 and 2 hold the sentences fixed and are the sensitive comparison.

and cost than API-served language models on service orchestration [14] and text annotation [15], but none tests it on neural decoding or against locally hosted scorers.

We test Jev as the rescoring stage of the published pipeline for participant T15. Every arm scores byte-identical cached candidate lists, so the rescorer is the only variable. We compare against the two 7B models the stage would otherwise use, under the published decoder setting and with the decoder’s weight re-tuned, and measure cost and latency.

Contributions.

- On real intracortical recordings, a typed-decision model matches two 7B rescorers: it is ahead of OPT-6.7b and Qwen2.5-7B under both tuning protocols, and the worst-case 95% bound puts it 0.2 points behind (Section 5.1).
- Posing the same typed decision to open general models fails: their deviations from the decoder’s top candidate select the truth 0.2–2.4% of the time against 26.6% for Jev (Section 5.2).
- Jev costs \$0.07 per thousand sentences with no local accelerator, and its API latency is dominated by transport rather than by the model (Section 5.3).

Cached lists, per-arm scores, latency logs and code are released.¹

2 Related work

Rescoring in speech neuroprostheses. Willett et al. [1] rescore with a large language model and report word error falling from 23.8% to 17.4%; Card et al. [2] use the same stage in a system that reaches single-digit error. Both use multi-billion-parameter models. We keep their pipeline and replace only the rescorer.

¹<https://github.com/gabrycina/how-much-language-model>

Rescoring in speech recognition. Automatic speech recognition interpolates the first-pass score with a language model’s likelihood [6] or trains the rescorer discriminatively [7, 8, 9]. Those methods apply here in principle, but they need training lists at a scale one participant’s 419-sentence development split cannot supply; in our experiments, minimum-word-error fine-tuning of GPT-2 small on 300 lists did not beat the decoder alone. Every arm here is therefore untuned, Jev included.

Listwise selection with language models. Listwise reranking asks a model to order or pick from a list in one pass. Its known failure is position bias, which permutation self-consistency [10] and order-invariant architectures [12] address. On an earlier decoder we applied permutation self-consistency to the open typed arms; neither beat the decoder’s own top candidate.

Typed-decision models. Li et al. [14] replace a language model with Jev for edge service orchestration and report 16–27% lower median latency and about 70% lower cost per correct completion than an API-served language model. Ibrahim and Zaki [15] evaluate decision models for text annotation and find Jev ahead of an open 0.6B model trained with the same objective. Both compare against generation over an API. Our baselines are scorers on a local GPU, which is the harder comparison for latency.

3 Background

For each attempted sentence the decoder emits a list $H = \{h_1, \dots, h_N\}$ ranked by

$$\ell(h) = \lambda a(h) + g(h), \quad (1)$$

where $a(h)$ is the acoustic score from the phoneme network, $g(h)$ the n -gram score and λ the acoustic weight, 0.3 in the published pipeline.

A rescorer assigns $r(h)$ to every candidate, and the system speaks

$$\hat{h} = \arg \max_{h \in H} (1 - \alpha) z_H[\ell](h) + \alpha z_H[r](h), \quad (2)$$

where z_H standardises a score to zero mean and unit variance across the list, so that rescorsers on different scales share one weight $\alpha \in [0, 1]$.

A per-candidate rescorer sets $r(h)$ to the length-normalised log-likelihood of h under a language model, one sequence per candidate. A typed rescorer receives the whole list once, with candidate h_i under label i , and returns a distribution p over labels; we set $r(h_i) = \log \max(p_i, 10^{-6})$.

4 Experimental setup

Data and candidate lists. We use the T15 recordings released with the Brain-to-Text benchmark [3, 2]: 256 electrodes over speech motor cortex, decoded by the authors’ pretrained recurrent network, unmodified. The published Kaldi decoder with an OpenWebText 3-gram produces 100-best lists for the 1,397 sentences of the benchmark’s validation split (oracle word error 2.5%, truth in the list 89%). A seeded split gives 419 development and 978 test sentences. Lists are cached, so every arm scores identical candidates.

Arm	published ($\lambda = 0.3$)			re-tuned λ		
	WER	Δ	p_{Holm}	(λ, α)	WER	Δ
No rescoring	8.09	—	—	(0.3, —)	8.09	—
OPT-6.7b	7.80	−0.29	0.070	(0.40, 0.45)	7.20	−0.89
Qwen2.5-7B	7.75	−0.34	0.070	(0.45, 0.45)	7.42	−0.67
Jev (typed)	7.46	−0.63	0.018	(2.0, 0.75)	6.93	−1.17

Table 1: Test word error (%) and paired difference from no rescoring. Under the re-tuned protocol all three arms separate from the baseline (Holm-adjusted $p \leq 0.002$). Published-protocol α : OPT 0.30, Qwen 0.45, Jev 0.90.

Arms. OPT-6.7b [4] and Qwen2.5-7B score every candidate on one NVIDIA L40S in bfloat16, each candidate as a standalone sentence, in batches of 32. Jev (`jev-latest`, September 2026) receives the list once through its API (Appendix A). As controls, Laya, an open 421M typed-decision model, receives the identical request, and OPT-6.7b and Qwen2.5-7B receive the same task and list as a text prompt, read from their probability for each label as the next token.

Tuning. Under the *published* protocol $\lambda = 0.3$ and α is tuned per arm over 21 values in $[0, 1]$. Under the *re-tuned* protocol λ is searched jointly over 17 values in $[0, 5]$. Both protocols select on the development split only and score the test split once. Without a rescorer the development split selects $\lambda = 0.3$ in both protocols.

Statistics. Word error is corpus-level Levenshtein distance over words (jiwer 4.0). Intervals are percentile bootstrap over sentences (2,000 resamples); comparisons are one-sided paired bootstraps (5,000) on the test split, each arm at its own tuned setting, with Holm correction over the three arm-versus-baseline tests within each protocol. We say Jev *matches* a 7B model when the 95% interval of their paired difference excludes a Jev deficit of more than 0.2 points. We fixed this bound after seeing the results (Section 5.4).

Cost and latency. Jev’s list price is \$0.042 per million input tokens with no charge for output; we read token counts from the API’s usage field on 140 random test lists. The GPU is an on-demand L40S at \$1.82 per hour. All arms were timed per list during scoring, from one data centre in Finland. We also timed Jev from London on 80 test lists, each paired with a two-option request, and recorded the provider’s `x-envoy-upstream-service-time` header, the time its proxy waits for the model service.

5 Results

5.1 Accuracy

Jev is the most accurate arm under both protocols (Figure 1, Table 1). At the published setting it reaches 7.46% against 7.80% for OPT-6.7b and 7.75% for Qwen2.5-7B, and it is the only arm that separates from no rescoring after correction. With λ re-tuned every arm improves and all three separate: Jev reaches 6.93%, OPT-6.7b 7.20% and Qwen2.5-7B 7.42%.

Paired against each 7B model, Jev is ahead in all four comparisons by 0.28–0.49 points (Table 2). Only the re-tuned comparison with Qwen2.5-7B reaches $p < 0.05$, and it does not

Protocol	Comparison	Δ WER	95% CI	p
published	Jev – OPT-6.7b	−0.34	[−0.80, +0.12]	0.082
published	Jev – Qwen2.5-7B	−0.29	[−0.78, +0.20]	0.129
re-tuned	Jev – OPT-6.7b	−0.28	[−0.72, +0.15]	0.112
re-tuned	Jev – Qwen2.5-7B	−0.49	[−0.98, −0.03]	0.019

Table 2: Jev against each 7B rescorer, paired on the test split. Negative favours Jev. The largest upper bound, +0.20, is the matching margin.

Typed arm	Labels used	Share on A	Truth picked	Truth picked ($\neq$ A)	WER
Jev	57 / 100	72%	67.9%	26.6%	7.46
Laya (421M)	44 / 100	69%	53.7%	2.3%	8.09
Qwen2.5-7B	31 / 100	28%	29.3%	2.4%	8.11
OPT-6.7b	13 / 100	35%	27.3%	0.2%	8.09
<i>Decoder’s top candidate</i>	—	—	68.0%	—	8.09

Table 3: The same typed task given to four models, all 1,397 lists. Label A is the decoder’s top candidate. “Truth picked ($\neq$ A)” conditions on the model choosing any other label. WER is on the test split at the published setting, α tuned; the three open arms stay at the baseline.

survive correction over the four tests (adjusted $p = 0.076$). The data therefore support equivalence rather than superiority: across both models and both protocols the 95% intervals exclude a Jev deficit larger than 0.20 points.

The tuned weights show how Jev is used. Its α is 0.90 at the published setting and 0.75 after re-tuning, against 0.30–0.45 for the 7B models, so the combination leans mostly on Jev. Re-tuning also moves Jev’s λ to 2.0, weighting the acoustic score far above the n -gram. A plausible reading is that Jev takes over the n -gram’s role of judging which candidate reads as English, leaving the decoder to supply acoustic evidence.

5.2 The typed decision needs a trained model

The gain comes from the model, not from the typed format. Given the same task, the three open models land on the baseline (Table 3); OPT-6.7b and Laya tune to $\alpha = 0$, so the combination discards them. All four typed arms place many answers on label A, the decoder’s own top candidate. What separates them is what happens when they deviate. Jev’s deviations select the truth 26.6% of the time; the open models’ deviations select it 0.2–2.4% of the time, close to noise. Re-tuning λ does not rescue them (8.09–8.37%). Candidates are listed in the decoder’s order, so position and decoder confidence coincide at the top; the open models’ answers track that position rather than the sentences.

5.3 Cost and latency

Jev reads a mean 1,631 input tokens per list (95% CI 1,398–1,868), which is \$0.069 per thousand sentences (Table 4). A GPU running OPT-6.7b scores 62,000 lists per hour when busy, \$0.029 per thousand, so a fully used GPU is cheaper per sentence. The GPU wins only above 43% utilisation, about 640,000 sentences a day. A single user speaks a small fraction of that, so a dedicated GPU sits mostly idle and costs \$43.68 a day, while Jev costs \$0.07 for a thousand

Arm	Cost (\$)		Latency per list (ms)	
	per 1k sentences	one user, per day	median	p95
Jev, API	0.069	0.07	282	346
OPT-6.7b, L40S	0.029*	43.68	27	134
Qwen2.5-7B, L40S	0.034*	43.68	32	158

Table 4: *At full GPU utilisation. “One user, per day” assumes 1,000 sentences a day and, for the 7B models, a dedicated on-demand GPU. Latency from the scoring runs, all 1,397 lists, one data centre in Finland; Jev’s includes the network.

sentences and needs no accelerator on the user’s side.

Jev answers in a median 282ms against 27ms for OPT-6.7b on the local GPU. The London measurement shows where the time goes. A full list takes a median 262ms end to end and a two-option request 253ms, so list size barely matters. The provider’s proxy reports a median 62ms for both (p95 102ms; 77ms for lists of 60 or more candidates), leaving about 194ms of transport between the client and the provider. The 62ms includes any queueing inside the provider, so it bounds the model’s own time from above. A co-located deployment would therefore approach 60–100ms, the same order as OPT-6.7b’s 27ms median and 134ms p95, but not below it. We could not test this, because the model runs only as a hosted service. At the roughly 32 words per minute reported for this participant [2], each sentence takes several seconds to attempt, so either figure adds little to the wait.

5.4 Limitations

- **One closed model.** Jev’s size and architecture are undisclosed, so the result shows that a typed-decision model can do this job, not why. We release its per-sentence label probabilities and settings so the analysis is reproducible, though the model is not. Hosted models change; our results are for `jev-latest` in September 2026.
- **Post-hoc margin.** We chose the 0.2-point matching bound after seeing the results; a confirmatory study should fix it in advance.
- **One participant, limited power.** T15 only, 978 test sentences; differences below about half a point are hard to resolve. 83 test sentences also occur, as text, in the phoneme network’s training split, a property of the benchmark; excluding them changes no conclusion.
- **Cloud dependence.** A hosted rescorer sends a user’s decoded speech to a third party and fails without a network. A clinical device would need a local or private deployment, whose latency we could not measure.
- **Prompt wording.** The typed prompt asks for the most fluent sentence, which could favour fluency over correctness. Jev picks the truth more often than the most fluent candidate when the two disagree, but we tested one wording.

6 Conclusion

On real intracortical recordings, one call to a typed-decision model rescores the candidate list as well as a 7B language model that scores every candidate: Jev is ahead of OPT-6.7b and

Qwen2.5-7B under both tuning protocols and at most 0.2 points behind at the 95% bound. It costs cents per thousand sentences and removes the GPU from the user’s side; its latency over the internet is transport-bound and, at the provider, the same order as a local 7B model.

Open general models given the same typed prompt fail, so the result depends on a model trained for the decision. The next step is an open typed-decision model of known size trained on this task, which would make both the cost and the latency claims testable on a patient’s own hardware.

A Prompt

Jev and Laya receive one request per list through the same interface, with candidates in the decoder’s order under labels A, B, and so on.

```
state = {"task": "A speech decoder produced these candidate sentences
              for what a person was trying to say. Exactly one is
              what they meant.",
        "candidates": {"A": <candidate 1>, "B": <candidate 2>, ...}}
questions = {"sentence": {
    "type": "choice",
    "instructions": "Which candidate is the sentence the person actually
                    meant? Judge by which reads as fluent, natural
                    English.",
    "criteria": {"A": <candidate 1>, "B": <candidate 2>, ...}}}
```

OPT-6.7b and Qwen2.5-7B receive the following text, wrapped in the model’s chat template where one exists and followed by **The answer is**; we read each label’s next-token probability.

A speech decoder produced these candidate sentences for what a person was trying to say. Exactly one is what they meant. Answer with the single label of the most fluent, natural English sentence.

A: <candidate 1>
 B: <candidate 2>
 ...

Answer:

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